

Payment Rails in Smart-Contract-as-a-Service (SCaaS) from BPMN Models

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Abstract

The adoption of blockchain-based smart contracts for the trade of goods and services promises greater transparency, automation, and trustlessness, but also raises challenges related to payment integration and modularity. While business analysts (BAs) can express business logic and control flow using BPMN and decision rules using DMN, payment tasks that involve concrete transfers (on-chain, off-chain, cross-chain, or hybrid) require careful implementation by developers due to platform-specific constraints and semantic richness. To address this separation of concerns, we introduce a methodology within the context of Smart Contract as a Service (SCaaS) that supports (1) identifying and mapping generic payment tasks in BPMN to pre-deployed payment smart contracts, (2) augmenting BPMN models with matching payment fragments from a pattern repository, and (3) automatically transforming the augmented models into smart contracts that invoke appropriate payment services. Our approach builds on prior work in automated BPMN-to-smart contract transformation using Discrete Event–Hierarchical State Machine (DE-HSM) multi-modal modeling to capture process semantics and nested transactions, while enabling payment service reuse, extensibility, and separation of concerns. We illustrate the methodology via representative use cases spanning conventional, DeFi, and cross-chain payments, and discuss implications for modular contract deployment and maintainability.

Introduction

Blockchains have attracted significant attention since the publication of the Bitcoin white paper in 2008 and the deployment of the Bitcoin blockchain in 2009. While public discourse has largely centered on cryptocurrency trading, blockchain technology has been widely embraced by business, research, and software engineering communities due to its key properties, including decentralised trust, immutability, availability, and transparency. These characteristics have motivated the rapid evolution of blockchain platforms, programming languages, and development tools for smart contracts. However, as with many emerging technologies, blockchains introduce new technical and engineering challenges that continue to hinder their widespread adoption.

At the infrastructure level, blockchain systems face persistent limitations related to scalability, transaction throughput, and cost. Eberhardt and Tai (2017) survey a range of techniques that shift functionality off-chain, classifying them into off-chain storage, off-chain computation, and hybrid solutions. Similarly, other studies investigate off-chain computation mechanisms that rely on cryptographic proofs to record verifiable results on-chain (Kosba et al., 2016; Luu et al., 2016). Another fundamental challenge stems from the cost of consensus protocols. Empirical assessments of blockchain capacity confirm that scalability remains a critical bottleneck, motivating research into alternative architectures, including sharding-based solutions (Buterin, 2017; Luu et al., 2016). Additional challenges include the oracle problem, transaction privacy, and interoperability between heterogeneous blockchain platforms, all of which have been extensively surveyed in prior literature (Atzei et al., 2017; Khan et al., 2020; Zheng et al., 2017).

Beyond infrastructure, the development and maintenance of smart contracts pose significant software engineering challenges. Smart contracts are subject to strict execution and deployment constraints, most notably immutability, since deployed contracts are permanently stored on the blockchain. As a result, bugs, vulnerabilities, or design flaws cannot be easily corrected after deployment, increasing the potential impact of development errors. These constraints heighten the need for rigorous software and security engineering practices and for development approaches that reduce complexity and error-proneness (Atzei et al., 2017). In this context, higher-level abstractions and automated generation techniques have gained increasing attention as a means of improving smart contract reliability and maintainability.

Smart contract development and maintenance have been widely studied, as reflected in comprehensive surveys [Atzei et al., 2017, Di Ciccio et al., 2020; Khan et al., 2020]. Several surveys highlight limitations in current smart

contract programming practices. Khan et al. (2020) distinguish between usage-oriented research and improvement-oriented research, where the latter focuses on tools and methods for improving smart contract development. Many improvement-oriented approaches rely on specialized programming languages, such as domain-specific or contract-oriented languages, that facilitate formal reasoning and verification. Despite their technical advantages, these languages have seen limited adoption in practice due to their perceived complexity and steep learning curve for developers.

Formal specification and verification techniques for smart contracts have also received considerable attention. A large-scale survey by Di Ciccio et al. (2020) classifies existing approaches according to their level of abstraction. Contract-level approaches model smart contract behavior using formalisms such as process algebras, state-transition systems, or set-based methods, while program-level approaches analyze artifacts such as abstract syntax trees, bytecode, control-flow graphs, or execution traces. While these techniques improve correctness and security, they generally assume the existence of a smart contract implementation and do not directly address the challenges of contract creation.

Model-Driven Engineering (MDE) approaches aim to address these challenges by generating smart contracts directly from higher-level models. A survey of MDE-based approaches highlights the use of modeling notations such as UML, BPMN, ERD, DEMO, ArchiMate, and EMF (Di Ciccio et al., 2020). Early work discusses the applicability of UML and BPMN models for blockchain-based applications without proposing concrete transformations (Wüst & Gervais, 2018), while other studies analyze blockchain technology using the business process management lifecycle and identify open research challenges (Viriyasitavat & Hoonsopon, 2019).

Among MDE approaches, BPMN-based approaches are particularly relevant due to BPMN’s widespread adoption in business process modeling and its suitability for representing multi-party workflows that can be enforced by smart contracts. Thus, researchers addressed the issue of generating smart contracts from BPMN models [Weber et al. 2016] using an approach that separates design-time transformation from run-time execution and introduces off-chain components for interaction with external systems, alongside on-chain components that enforce process choreography. This work led to the development of the Caterpillar (López-Pintado et al., 2017) and Lorikeet (López-Pintado et al., 2019) frameworks, which serve as reference platforms for BPMN-driven smart contract execution. Subsequent research, which extended these platforms in several directions, will be reviewed in detail in a later section of this paper when discussing related work.

As in other approaches to transforming BPMN models to smart contracts, we also start with the Business Analyst (BA) preparing a model of the business processes occurring in the trading activity using BPMN. The BA also describes the business logic using the Decision Modeling Notation (DMN). However, in comparison to other research in this space, we use a unique approach to transformation of the BPMN models to smart contracts that uses multi-modal modeling and provides support for *multi-step nested transactions privacy and security*, and *support for Smart Contract Lifecycle* (Bodorik, Liu, Jutla, 2023; Liu, Bodorik, Jutla, 2024; Liu, Bodorik, Jutla, 2025; Dalhousie Blockchain Lab, 2024).

Despite these advances, payment integration remains a challenge: it is unreasonable to expect a BA to specify low-level payment implementation details in the target programming language (e.g., Solidity on Ethereum). Different trade applications may require disparate payment methods (bank rails, crypto tokens, escrow, cross-chain bridges), each with distinct semantics and compliance constraints. Furthermore, modern systems increasingly hybridize on-chain and off-chain components, demanding orchestration logic that ensures atomicity and correct sequencing across disparate execution contexts.

Across all payment categories, two properties are especially relevant to the BPMN-to-smart contract transformation problem:

1. *Semantic richness* — payments involve amounts, assets (currencies or tokens), participants, timing constraints, and compliance conditions, all of which must be respected by execution logic.
2. *Implementation diversity* — the actual execution semantics vary widely depending on the payment rail (traditional banking, DeFi protocols, escrow logic, wrapped assets, or cross-chain protocols), even though BAs typically model payments as abstract tasks.

This motivates the central abstraction of SCaaS: BAs model payments generically, while developers produce concrete payment smart contract services that can be pre-deployed and invoked by automatically generated orchestration of process execution from BPMN models.

In our previous work, we laid the foundation for automated generation of smart contracts from BPMN models using Discrete Event–Hierarchical State Machine (DE-HSM) multi-modal modeling, which captures BPMN control flow as a network of interacting FSMs enriched with discrete event semantics. The system transforms BPMN models

into DE-HSM representations and then into contract methods, enabling off-chain or sidechain execution to reduce cost and increase privacy. (Bodorik, Liu, Jutla, 2023)

Subsequently, we extended this pipeline to support nested collaborative transactions (TABS+), identifying BPMN sub-patterns suitable for complex transactional coordination among multiple actors and generating contract mechanisms to enforce transactional properties automatically. (Bodorik, Liu, Jutla, 2024)

More recently, we introduced TABS+R, a mechanism to repair and upgrade generated contracts when real-world conditions diverge from modeled assumptions, by isolating problematic BPMN fragments, amending them, and regenerating consistent contract logic. (Bodorik, Liu, Jutla, 2025)

While these approaches substantially reduce developer effort and increase correctness, they do not fully address payment rail abstraction and reuse. In particular, they do not provide a systematic way to separate payment implementation concerns from process logic, nor the ability to reuse pre-deployed payment services across multiple trade applications.

1.1 Objectives

Within the context of generating Smart Contract as a Service (SCaaS) from BPMN models, this paper aims to:

1. Enable software developers to *prepare specific payment (smart contract services)* in advance, targeting various payment rails (on-chain crypto, off-chain bank transfers, cross-chain rails).
2. Allow business analysts to model trade activities (goods/services) in BPMN using a *generic representation of payment tasks*.
3. Design a transformation process that — given a BA’s BPMN model — *identifies generic payment tasks*, matches them to specific payment smart contract services via a *BPMN-fragment repository*, augments the BPMN model accordingly, and generates a smart contract that orchestrates the trade activity while invoking appropriate payment services.

1.2 Contributions

This paper makes the following specific contributions:

- We propose a *separation of concerns* between trade-activity modeling (performed by BAs) and payment service implementation (performed by developers), enabling generic payment task modeling in BPMN.
- We design a *BPMN-fragment repository*, a library of payment BPMN patterns annotated with metadata (inputs, outputs, state dependencies) to enable automated matching based on semantic criteria.
- We describe a transformation process (*SCaaS tool*) that generates smart contracts from augmented BPMN models, invoking pre-deployed payment service methods rather than re-implementing them, thereby enhancing reuse and modular deployment.
- We demonstrate how this methodology supports *extensibility and reuse*: payment services can be shared across multiple trade processes, and new payment methods can be integrated by simply adding new BPMN fragments and deploying their associated smart contracts, without modifying existing trade activity models.
- We support heterogeneous payment scenarios, including off-chain conventional payments, on-chain crypto payments, and cross-chain mechanisms, illustrating adaptability to diverse business needs.

1.3 Outline

The remainder of this paper is organized as follows. Section 2 provides background on conventional and crypto payment methods, introduces BPMN and DMN modeling and their use in transformation pipelines, and overviews our SCaaS approach from BPMN and DMN models. Section 3 presents our approach to integrating payment services into SCaaS workflows, including repository design and smart contract generation, illustrated through a representative use case covering on-chain, off-chain, and cross-chain payments. Section 4 discusses reuse and extensibility by presenting an additional use case of creating a payment service that utilizes payments with settlement netting while re-using other, already existing, payment services and showing how the repository supports integration of services. Section 5 reviews related work, situating our contributions relative to existing BPMN-to-smart contract transformation research. Finally, Section 6 provides a summary and conclusions, and outlines directions for future work.

2. Background

This section surveys foundational concepts underlying the generation of smart contracts from business process models and motivates the Smart Contract as a Service (SCaaS) paradigm with explicit support for payment rail abstraction. We first characterize payment methods across conventional and blockchain-based ecosystems, then introduce BPMN and DMN as modeling standards for business processes and decisions and finally review our approach to transforming BPMN models into executable smart contracts.

2.1 Payment Methods in Conventional and Blockchain Ecosystems

Payments represent the transfer of economic value between parties and differ substantially in semantics, execution guarantees, and compliance obligations depending on the underlying rail. These differences are especially relevant when payment logic is orchestrated by smart contracts generated from abstract process models.

2.1.1 Conventional (Off-Chain) Payments

Conventional payment systems operate over established financial infrastructures such as Automated Clearing House (ACH) transfers, wire payments, card networks, Real-Time Gross Settlement (RTGS) systems, and correspondent banking arrangements. These rails rely on intermediaries, including banks, clearing houses, and payment processors, to execute settlement, manage liquidity, and enforce regulatory requirements such as Anti-Money Laundering (AML) and Know Your Customer (KYC) checks. Payment instructions and confirmations are commonly exchanged using standardized messaging frameworks, notably SWIFT MT and MX formats ((BIS, 2012; SWIFT, 2020).

Settlement in conventional systems is typically irreversible once completed, and dispute resolution is handled through institutional or legal processes external to the execution infrastructure. While compliance rules and exception-handling logic can be encoded in smart contracts, the determination of compliance and exception conditions typically depends on off-chain data, institutional authority, and human judgment. As a result, compliance evaluation and exception recognition cannot be performed natively on-chain and must be delegated to external services or oracles when integrating conventional payment rails with smart contracts. (Narayanan et al., 2016)

2.1.2 Blockchain-Based Payments

Blockchain-based payment mechanisms enable programmable value transfer on distributed ledgers. These mechanisms can be classified based on ledger locality and the nature of inter-ledger interaction.

On-Mainchain Payments

On-mainchain payments occur when both the requesting smart contract and the payment execution logic reside on the same blockchain. Transfers of native cryptocurrencies or tokenized assets are atomic, transparent, and self-verifying due to blockchain consensus. Such payments support advanced features such as conditional transfers, escrow, and bulk payments with netting, all enforced directly by smart contract logic (Buterin, 2014; Szabo, 1997). Regulatory compliance is typically layered via external oracles or application-level controls.

Cross-Chain Native Payments

Cross-chain native payments transfer value between distinct blockchains while preserving asset equivalence. Typical approaches rely on burning or locking assets on the source chain and minting or releasing corresponding assets on the destination chain via bridges or cryptographic interoperability protocols. Correct execution requires atomicity and state consistency across heterogeneous ledgers (Narayanan et al., 2022; Zamyatin et al., 2021). However, the most difficult issue is how to preserve asset equivalence, particularly if the asset value depends on its demand and may change rapidly, such as the value of publicly traded stocks or prices of some of the crypto-coins.

Third-Party Cross-Ledger Payments

In third-party cross-ledger payments, a trade-activity smart contract deployed on one blockchain invokes a payment service hosted on a different blockchain. The 3rd-party payment service executes the transfer using its native asset, such as crypto-coins stablecoins USDT or USDC, or emerging Central Bank Digital Currencies (CBDC), and returns a success or failure response. As the payment service is based on payment request/reply message exchange, this interaction resembles and is functionally similar to off-chain payment services from the caller's perspective (Wood, 2016; Garrat, 2020).

Clearly, the 3rd-party payment services, is a special subset of the Cross-Chain Native payments, in which both the payer and payee have accounts on the 3rd-party payment service blockchain, and both parties agree on a specific payment value expressed in native crypto-coins of the payment service blockchain. It is expected that when CBDC

payment services are established, they will become the dominant payment services used most often and become the corner stone of web3 payment services.

Hybrid Payments

Hybrid payment systems combine off-chain, on-mainchain, and cross-chain components using wrapped assets, bridges, or interoperability frameworks. While offering flexibility, these systems introduce additional orchestration complexity, making them a natural fit for programmable controllers implemented as smart contracts. (Belchior et al., 2021)

Across all payment categories, two properties are particularly salient for BPMN-to-smart-contract transformation: semantic richness, encompassing amount, asset type, timing constraints, participating parties, and compliance conditions; and implementation diversity, whereby identical business-level payment tasks map to radically different execution semantics depending on the rail (Pahl, El Ioini, & Helmer 2019; Weber, 2016). These characteristics motivate a service-oriented abstraction in SCaaS, where business analysts model payments generically and developers implement reusable payment services.

2.2 BPMN, DMN, and FEEL

Model-driven smart-contract generation depends on modeling standards that balance expressiveness with accessibility for business stakeholders.

2.2.1 Business Process Model and Notation (BPMN)

Business Process Model and Notation (BPMN) is an ISO-standard graphical language for specifying business processes involving tasks, events, gateways, message flows, and collaborations (Object Management Group, 2013). BPMN enables business analysts to express control flow, parallelism, and participant interaction in an intuitive manner. In spite of its rich expressiveness in terms of process-flow modeling, it was designed to act as a "common language" that bridges the communication gap between initial business process design and final technical implementation. Thus, it was specifically designed to be understood by three primary groups of business stakeholders, namely: *Business Analysts* who use the notation to create the initial drafts and refinements of business processes; *Technical Developers* who are responsible for implementing the technology and software components that will execute those processes; and *Business Managers* who use the diagrams to monitor, manage, and make strategic decisions based on the processes.

Importantly, BPMN is technology-agnostic: tasks describe business intent without embedding execution-level semantics, which facilitates transformation into multiple target platforms but requires additional interpretation during code generation.

2.2.2 Decision Model and Notation (DMN) and FEEL

Decision Model and Notation (DMN) complements BPMN by providing a standardized approach for modeling business decisions and rules (Object Management Group, 2023). DMN decision logic is expressed using the Friendly Enough Expression Language (FEEL), a declarative, human-readable language designed to be both understandable by business users and executable by machines. BPMN tasks may invoke DMN decisions to guide control flow, enabling separation between process structure and decision logic in model-driven transformations. (Object Management Group, 2013; Object Management Group, 2023)

2.3 BPMN-to-Smart-Contract Transformations

A substantial body of research explores transforming BPMN models into executable artifacts, including smart contracts. These approaches commonly treat BPMN as a high-level specification of process behavior, which is mapped to executable representations such as finite state machines or choreographed services (e.g., López-Pintado et al., 2017; López-Pintado et al., 2019; Weber, 2016). In the following, we shall briefly overview our approach to transforming BPMN models into smart contracts from BPMN models, an approach to which we refer as SCaaS, and its tool, SCaaS tool, which we called in our previous papers as the TABS tool, and later TABS+, and TABS+R, as the tool evolved (Dalhousie Blockchain Lab, 2024). In a later section, called Related Work, we situate our work within the context of research on transformation of BPMN models into smart contracts. Our primary differentiator from other work on this topic is that we have developed a sequence of multi-modal modeling approaches that employ Discrete Event–Hierarchical State Machine (DE-HSM) representations to bridge BPMN models and smart contracts.

The *TABS approach* and its TABS tool convert BPMN models into smart contracts through an intermediate DE-HSM representation that captures both control flow and execution state. Logic-Structured Independent (LSI) subgraphs of BPMN models are identified and transformed into executable contract methods, enabling modular generation and optional sidechain execution for cost and privacy optimization (Bodorik, Liu, Jutla, 2023).

TABS+, the next version of the tool, extends the TABS framework to support multi-step collaborative transactions involving multiple participants. By identifying transactional scopes within BPMN models, *TABS+* generates smart contract logic that enforces governance rules, checkpoints, and coordination semantics across actors (Liu, Bodorik, Jutla, 2024).

TABS+R addressed the need to handle unanticipated events that arise in trade. While automated BPMN-to-smart-contract generation and nested transaction support address many design-time challenges, real-world processes inevitably evolve or encounter unanticipated runtime conditions. To improve usability and robustness, we introduced *TABS+R* tool, extending the transformation pipeline in two key directions:

(i) First, *business analyst-driven logic* is enabled through integration with Decision Model and Notation (DMN). Business analysts can specify conditional decision logic using DMN and FEEL directly within BPMN models, and this logic is automatically incorporated into the generated smart contracts. As a result, domain experts can modify process behavior and decision rules without direct involvement from smart contract developers, significantly improving accessibility and maintainability.

(ii) Second, *repair and upgrade support* is provided to address runtime failures and process evolution. *TABS+R* identifies the innermost failing transactional BPMN fragment, allowing the modeller to revise the corresponding BPMN/DMN logic and regenerate the smart contract. The regeneration process preserves execution consistency by systematically reconstructing contract logic and transaction mechanisms, enabling controlled upgrades while maintaining state continuity.

Together, these extensions make the BPMN-to-smart-contract pipeline resilient to change and failure, supporting end-to-end automation from modeling through deployment, repair, and evolution in practical business settings.

However, despite their strengths, TABS and *TABS+* assume that BPMN tasks are either directly implementable or require business logic. Payment tasks, however, often involve external services, heterogeneous rails, and reusable protocols. These limitations motivate the SCaaS abstraction, which separates payment implementation from process logic and enables payment services to be invoked as modular components during smart-contract generation.

3. Making Payments - Service Modeling, Deployment, and Use

When considering payment services, there are two perspectives, payer's and payee. To receive a payment, the payee needs to provide information needed to make the payment using a specific payment method. For instance, the instructions may state that the payment is to be made to a specific bank account, while also providing information that depends on whether the transfer is inter or intra-bank transfer and whether it is for a cross-border payment. For crypto-payment method, the payee needs to identify the crypto token/currency and other information required, such as invoice or payment ID that identifies the goods/services for which the payment is made.

We first overview, in the section 3.1, the overall process of making payments, how they are modeled, stored, and deployed within the SCaaS pipeline, and how generic payment tasks in business process models are systematically mapped to specific pre-deployed payment smart contracts. Next, we detail how payment service smart contracts are prepared and deployed (Section 3.2). We then present the structure and use of a *payment service repository*, which mediates the mapping between generic BPMN payment tasks and concrete services (Section 3.3). Finally, we describe the end-to-end transformation process that generates trade smart contracts from augmented BPMN models while invoking repository-selected payment services (Section 3.4). The exposition describes how the interpretive transformation framework, previously established for BPMN-to-smart contract synthesis using DE-HSM multi-modal modeling, was extended to support payment services.

3.1 Overview of Payment Services in SCaaS

Our approach extends the SCaaS paradigm by decoupling business-level payment modeling from payment implementation, enabling business analysts (BAs) to express payments *generically* while developers supply executable services. In SCaaS, BPMN models are transformed into Discrete Event-Hierarchical State Machine (DE-HSM) models that preserve BPMN control flow semantics and facilitate automatic smart contract generation. Generic payment actions in BPMN are modeled as simple task elements annotated with necessary data objects, such as payee, invoice ID, amount, currency, and other domain-specific parameters. These annotations capture *semantic inputs* flowing into the payment task, providing the metadata needed to later select appropriate payment services. (Bodorik, Liu, Jutla, 2023).

Unlike task elements that represent on-chain logic or business interactions (captured via BPMN tasks and DMN decision rules), a generic payment task does not itself encode execution semantics. Instead, at transformation time, the SCaaS tool consults a payment service repository to identify which concrete payment smart contract (pre-deployed by developers) matches the semantic characteristics of the generic task. Once selected, the BPMN model is *augmented* with the BPMN fragment corresponding to that payment service and transformed into a composite smart contract that invokes the selected service rather than re-implementing payment logic. This separation supports reuse, modular deployment, and extensibility.

3.2 Preparation and Deployment of Payment Service Smart Contracts

Payment service smart contracts, whether payables or receivables, are developed by software engineers as reusable SCaaS services targeting specific payment rails (e.g., conventional banking via off-chain APIs, on-chain native token transfers, or cross-chain services via bridges/oracles). Each service originates from a payment BPMN fragment model that describes the service's control and data flow, including required inputs and outputs.

The developer first prepares the BPMN fragment to model the payment. Following the standard SCaaS transformation, the BPMN fragment representing the payment is processed into an equivalent DE-HSM model and then automatically compiled into smart contract methods. As with other BPMN-to-smart contract transformations, concurrency and control flow are represented via DE queues and state machines to preserve semantic correspondence between BPMN and contract logic. Once synthesized, the payment service contracts are deployed to designated blockchain environments (e.g., Ethereum, a DeFi-capable chain, or a sidechain configured for cross-chain interactions). Deployment metadata, including method signatures, required parameters, compliance constraints, and supported rails, is recorded in the payment repository for later retrieval.

Conceptually, payment services fall into the following categories based on their execution context. The first classification is whether the payment services are for (a) *making payments* and (b) *receiving payments*, which are also referred to as *payables* and *receivables*, respectively. Both payables and receivable services are further classified as follows:

- i. *On-chain native payments* – Simple transfers of native tokens (e.g., ETH) between accounts on the same blockchain. These payments are *atomic* and *synchronous*, and smart contract invocation typically triggers immediate settlement within the same transaction context.
- ii. *Cross-chain crypto payments* – Payments facilitated across distinct blockchains using bridges or oracle mechanisms. These involve auxiliary/helper contracts on each chain to manage requests and callbacks, resulting in *asynchronous* execution and eventual consistency once the cross-chain protocol finalizes.
- iii. *Off-chain conventional payments* – Payments mediated by external financial systems (e.g., bank transfers) via HTTP APIs or web services. These are inherently asynchronous and non-atomic, requiring a two-phase interaction pattern (request and callback) to reconcile success or failure.

Developers must ensure that each deployed payment service complies with required protocols, including cross-chain communication interfaces or third-party API integration for off-chain rails.

3.3 Payment Service Repository

To mediate between *generic BPMN payment tasks* and *specific deployed services*, we maintain a structured *payment service repository*. Each entry in the repository corresponds to a deployed payment contract and includes (Buterin, V., 2014; Gudgeon et al. (2020), WWW Consortium (2017)).

- **Semantic descriptors** – Formal metadata capturing payment type (payable/receivable, on-chain, off-chain, cross-chain), expected currency or token, blockchain network identifiers, and compliance requirements.
- **Method signatures** – Information about required method parameters (e.g., sender/receiver identifiers, amounts, currency codes), event callbacks, and expected outputs.
- **Compliance and operational metadata** – KYC/KYB/AML information requirements, netting details, fee handling semantics, and any advanced operational constraints necessary for service invocation.

Repository entries are indexed to support structured search based on BPMN task metadata, with the first index determining whether the service is for making or receiving a payment. When SCaaS encounters a generic payment task during transformation, it uses the task's annotated inputs to filter candidate services that satisfy the semantic

criteria (e.g., payable vs. receivable, correct currency, supported rail, required data fields). Partial matches prompt interactive refinement, and multiple matches allow user selection among alternatives. Once resolved, the system links the BPMN payment task to the selected service's BPMN fragment and embeds invocation logic in the augmented model.

This repository mechanism supports repository evolution because new services can be inserted when deployed, and usage logs can be stored to identify commonly selected payment rails for particular business contexts.

3.4 Transforming the Augmented BPMN Model into Smart Contracts

Once payment services are selected and integrated into the BPMN model, the SCaaS transformation pipeline proceeds as follows:

1. *BPMN augmentation*: Generic payment task elements are replaced or extended with repository-selected BPMN fragments that represent concrete payment service invocation logic, preserving original sequencing and exception handling.
2. *DE-HSM conversion*: The augmented BPMN model — now containing both business logic and payment semantics — is transformed to a DE-HSM multi-modal model. The DE layer captures event sequencing and concurrency, while HSM sub-models encode task execution semantics (both business and payment).
3. *Smart contract synthesis*: Using the TABS transformation mechanism, each DE-HSM sub-model is translated into smart contract methods. Payment service invocation is realized by embedding calls to the deployed payment smart contracts (identified via repository metadata), using appropriate inter-contract communication patterns and callback interfaces where necessary.
4. *Deployment and runtime artifacts*: The generated smart contracts are deployed to the target blockchain(s), and auxiliary APIs are produced to interface with the application layer. Integration with bridge/oracle services (for cross-chain rails) and off-chain callback handlers (for conventional payments) is managed via event listeners and workflow coordination logic.

Integrating payment services at transformation time preserves the developer's prior work and ensures that payments occur according to pre-tested semantics, rather than regenerating payment code for each trade process. This enhances modularity, reduces redundancy, and aligns with the SCaaS objective of factorizing reusable service logic from business process orchestration. In the following subsection, we describe the transformation process in a use case.

It should be noted that in the rest of the paper, when we refer to the payment services, we refer to both payable and receivable services. When we wish to differentiate, we shall make explicit references to payment-making service or payment-receiving services, or to payable or receivable services, respectively.

Also note that the repository stores the BPMN fragment used for transformation into a payment service smart contract in addition to the other information, such as whether it is a payable or receivable service, whether off-chain or some other type of service. The BPMN fragment model is included as it is necessary for interpreting status-events emitted by smart contracts. The following subsection will provide examples to clarify.

3.5 End-to-End Process for the Use Case

To show the end-to-end process for generating a blockchain application, from a BPMN model that includes payments, we first describe a use case utilized for exposition purposes and overview our experimentation environment. We then describe creation of payment services smart contracts by the developer and overview the process of transforming the BPMN model of the trade activity into smart contracts that invoke the payment services prepared by the software developer. Finally, we describe the execution and monitoring of the smart contract execution.

3.5.1 Sample Use Case

The BPMN model for the sample use case is shown in [Figure 1](#). It shows a BPMN model that represents a sale of a large product, such as a combine harvester. Modeling is carried out by a Business Analyst (BA) who is assumed to be proficient in BPMN and DMN modeling, including the use of the FEEL language for decision logic. Additionally, we assume that the BA is familiar with JavaScript Object Notation (JSON), which is used to describe the flow of information throughout the computation process, as will be detailed later.

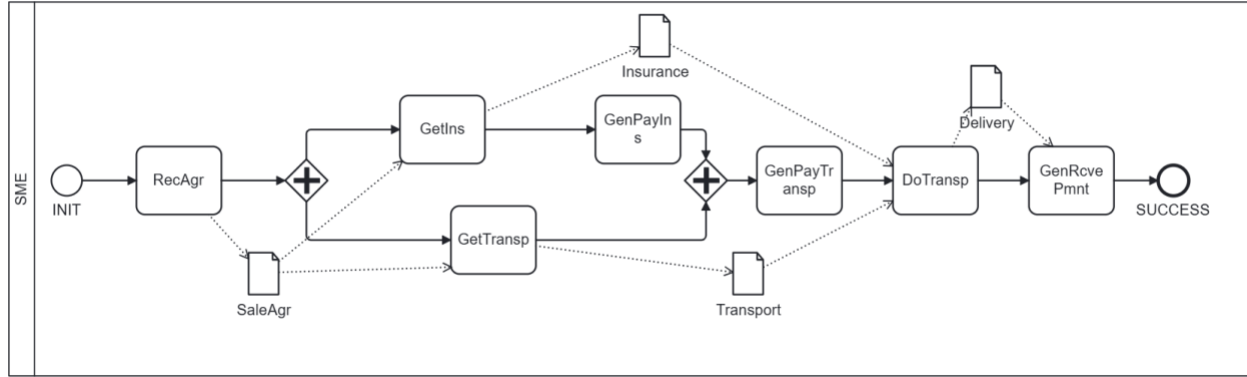


Figure 1. Sample Use Case: BPMN Model with Generic Payments *GenPayIns*, *GenPayTransp*, and *GenRcvePmnt*

In Figure 1, the first task, *RecAgr*, involves receiving an already approved purchase agreement, such as an approved purchase order. The agreement is represented by a BPMN associated data element, *SalesAgr*. The BA prepares the requests for insurance and transport represented by the task elements associated *GetIns* and *GetTransp*, while providing the task elements information via the associated document *SalesAgr*. The task *GetIns* obtains the insurance contract, represented by the *Insurance* associated document, which is followed by the task element *GenPayIns* representing the generic payment for the insurance that was obtained.

The task *GetTransp* obtains a contract from a transporter for the product transport, with the contract being described in the *Transport* associated document. Obtaining and paying for insurance and obtaining the transporter can be executed concurrently, as indicated by the split parallel gateway (diamond shape with a plus sign). Once both insurance and transport contracts are obtained, the payment to the transporter is made as represented by the generic payment task *GenPayTransp*. The product is then received by the buyer, which is recorded in the *Delivery* associated document that is forwarded to the final task, *GenRcvePmnt*, that receives the payment from the buyer.

It should be noted that the use case was utilized as a sample use case in our previous publications in (Liu, Bodorik, Jutla, 2024b; Liu, Bodorik, Jutla, 2025c). In (Liu, Bodorik, Jutla, 2025c), we described how a BA, working within the context of an SME, creates a BPMN model to track activities, document flows, and express the business logic decisions of BPMN task elements using DMN modeling. However, as we are dealing with payments, we extended this use case for the purposes of this paper with the following payments:

- Before the insurance can take place, it must be paid, which is represented by the BPMN task element *GenPayIns*.
- Payment for the transport is performed once the transport is completed. The payment is represented by *GenPayTransp* task element.
- Once the product is received, the buyer pays for the product, which is represented by the *GenRcvePmnt* BPMN task element.

Furthermore, the naming, or labeling, of the BPMN task elements is such that any task name that starts with the substring “*GenPay*” represents a payable payment service, wherein the rest of the string name must ensure the name’s uniqueness. Similarly, any BPMN task element with a name/label that starts with a substring “*GenRcve*” represents a general receivable payment service.

3.5.2 Execution Environment

(* To be discussed / determined about what needs to be described for experimentation and where presented***)**

We use three cloud servers provided by DigitalOcean (DigitalOcean, n.d.). Each server is equipped with two CPUs, 4 GB of memory, and 80 GB of disk space, and runs the Ubuntu operating system. Ganache-CLI was installed, configured, and executed on each server with different parameters to emulate a realistic deployment environment. The mainchain blockchain was configured with varying parameters for block time, number of endorsers, and block size, using values close to those of the public Ethereum blockchain. The sidechain networks were configured using Quorum for Ethereum smart contracts and Microfab for Hyperledger chaincode.

In addition, the TABS tool allows testers to deploy smart contracts on the public Ethereum mainnet and the Ropsten test network, provided that sufficient cryptocurrency is available on those blockchains. Furthermore, each server also functions as an IPFS node within a preconfigured private IPFS cluster, ensuring that each participant has dedicated access to its corresponding IPFS node and storage space.

The evaluation process for a use case begins with BPMN modeling. Upon completion of the modeling phase, TABS automatically transforms the BPMN model into a DE-HSM model during the design phase. The modeler then provides the implementation code for the tasks defined in the BPMN elements. The evaluation process for a use case begins with BPMN modeling. Upon completion of the modeling phase, TABS automatically transforms the BPMN model into a DE-HSM model during the design phase. The modeler then provides the implementation code for the tasks defined in the BPMN elements. The resulting DE-HSM model is presented to the modeler who is asked to provide information on which deployed services are to be used for the generic payment services present in the BPMN model.

After the compiled smart contracts are deployed on the blockchain networks, the modeler selects the evaluation process to be used during the runtime phase. During execution, the modeler can observe the invocation of smart contract methods. The TABS tool supports evaluation of the deployed model using an input stream that simulates expected inputs from a decentralized application (DApp), as they would be submitted through the DApp interface. Performance metrics include the delay per smart contract invocation, the completion time of a sub-model execution, and the total execution time of the DApp, measured from the start event to the end event in the BPMN diagram.

(***** Based on Elsevier Journal 1 or 2 *** above ... to be removed

3.5.3 Payment Services and Repository

When an invoice is submitted for payment of some goods or services, also submitted with the invoice are instructions on how to make the payment that specifies information that needs to be provided for the payment execution. If the required payment does not yet have a payment service in the repository, it must be prepared by a software developer using the instructions for payments provided by the payee.

As described in the subsection 3.2, the developer first models the BPMN fragment that represents the payment service and then uses SCaaS tool to transform the model into the methods of a smart contract and APIs. We assume that instructions for the insurance payment, represented by the BPMN task element *GenPayIns*, indicate that the 3rd-party crypto-payment is to be made using the *USDXCrypto-StCn* stable-coin blockchain, while the payment instructions for the *GenPayTransp* indicate that the payment is to be made via an off-chain bank transfer. Finally, the task element *GenRcvePmnt* represents the payment receivable service, which is to be performed by the on-mainchain crypto payment, i.e., it is to be made by the crypto-coin of the mainchain on which the trading activity is deployed, and hence the payment service and the trade activity smart contracts share the ledger.

Here we shall concentrate on the basic payment services accomplished by a payment service that does not re-use other payment services. The payment methods of the sample use case are created by the software developer, but once created, they can be re-used as shall be demonstrated in the next section.

From the perspective of the BPMN semantics, once the flow of control arrives at the BPMN task element representing a payment service, it completes its execution before the flow of control leaves the task element via an exit sequence flow represented by an outgoing arrow from the task element. This BPMN semantics applies to all payments, regardless of whether the payments on the mainchain, cross-chain, or off-chain. However, from the execution architecture point of view, there are difference, depending on the implementation of the payment services.

If the payment services and the trade activity share the ledger, then the payment is accomplished by an atomic transfer of crypto units. However, the cross-chain and off-chain payments require the use of a helper smart contract or an oracle to provide a bridge to the crypto-payment blockchain: The helper smart contract method interacts with the target crypto-payment blockchain or the off-chain service requesting it to make the payment and then receives from the payment service indication of the success/failure. The helper smart contract, upon receiving the reply from the payment blockchain, informs the trade activity whether the payment succeed or failed. The BPMN fragment model is shown in [Figure 2](#). The trade activity smart contract invokes the helper smart contract method that provides a bridge to support invocation of the helper smart contract on the crypto-payment blockchain.

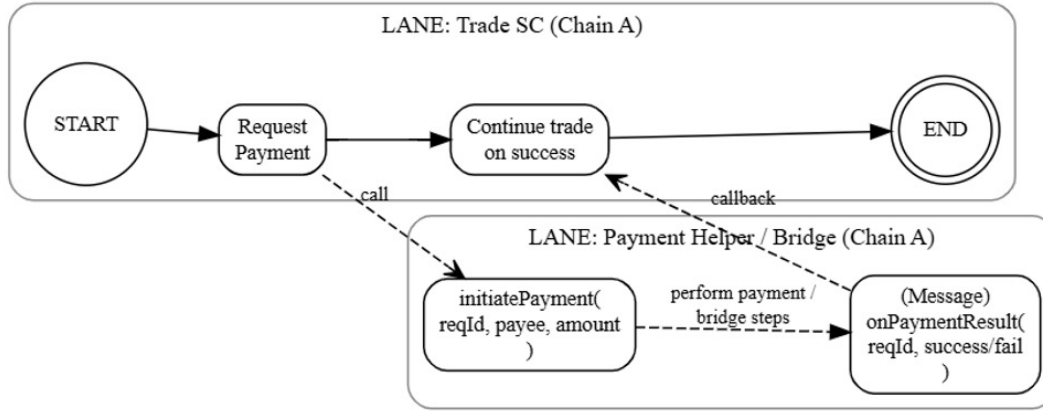


Figure 2. Request-reply interaction with the helper smart contract for cross-chain payment

Once the payment is performed, the helper smart contract on the payment bridge interacts with the helper smart contract on the blockchain with the trade activity smart contract. Thus, in the figure, the helper smart contract is shown to be included in a different swimlane, which is a BPMN term, represented in the figure as a rectangle encompassing the BPMN model elements representing the execution of the helper smart contract. The dashed arrow labelled “Perform Payment/Bridge Steps” is used to represent payment activities executed on the either the payment blockchains or off chain. Completion of such activities concludes with communicating the success/failure to the helper smart contract executing on the blockchain with the trade activity smart contract. Recall that for the cross-chain payment service, the developer must also prepare a smart contract on both blockchain involved, the trade activity and payment blockchains.

The role of this helper contract is not to execute payment logic, but rather to (i) register payment intents, (ii) record attestations of payment outcomes, and (iii) emit verifiable events that allow the on-chain execution to progress. The use of a helper contract provides a uniform abstraction for off-chain and cross-chain payment services, including crypto-based payment rails (e.g., sidechain transfers, cross-chain payments) and conventional payment services (e.g., fiat payment processors). In this unified model, the main smart contract interacts exclusively with the helper contract, while the helper contract interfaces—directly or indirectly—with off-chain services.

However, for purely conventional off-chain payment methods, where no blockchain-resident artifact is required beyond confirmation of payment completion, the helper contract may be omitted. In such cases, the off-chain payment service interacts with the main smart contract through a minimal callback mechanism that submits an attestation of payment completion. Conceptually, this corresponds to a degenerate instance of the helper-contract pattern. However, we do not support such an option.

3.5.3 Transforming BPMN Model and Execution

(* ** From Tabs+R Elsevier J2 paper *** **)** In our previous work (Bodorik, Liu, Jutla, 2023), we demonstrated the feasibility of transforming BPMN models into smart contracts using the *TABS tool*. We subsequently introduced the concept of *nested trade transactions* in smart contracts to support complex collaborative activities that exceed the capabilities of native blockchain transactions (Liu, Bodorik, Jutla, 2024a). To this end, we extended the TABS tool into *TABS+*, enabling the automatic generation of smart contracts that implement nested trade transactions while also providing mechanisms to enforce transactional properties. In later work (Liu, Bodorik, Jutla, 2025), we further extended the tool into *TABS+R*, which supports contract repair when external conditions prevent the completion of the originally generated contract. In the present paper, we describe a further extension of this toolchain with explicit support for payment rails, resulting in the *SCaaS tool*.

It should be noted that the SCaaS tool is configured primarily for research, experimentation, and testing. Accordingly, it includes features and processing steps that are not intended for production environments, such as the ability to step through individual transformation phases and inspect intermediate inputs and outputs. The tool also supports the issuance and execution of multiple transactions and enables the measurement of execution delays by capturing timestamps at various points during processing.

The SCaaS tool utilizes the Camunda platform (Camunda, 2025) for creating BPMN models in accordance with the BPMN specifications (OMG 2013, 2023). BPMN models are stored in XML format. **Figures 3** shows a screenshot of the SCaaS tool during BPMN model creation for the example application. Once a BPMN model is

saved in XML form, it is transformed through a sequence of steps controlled via tabs in the tool's user interface. The initial tabs support BPMN modeling, while subsequent tabs guide the user through the transformation process, including smart contract generation, deployment, and execution control.

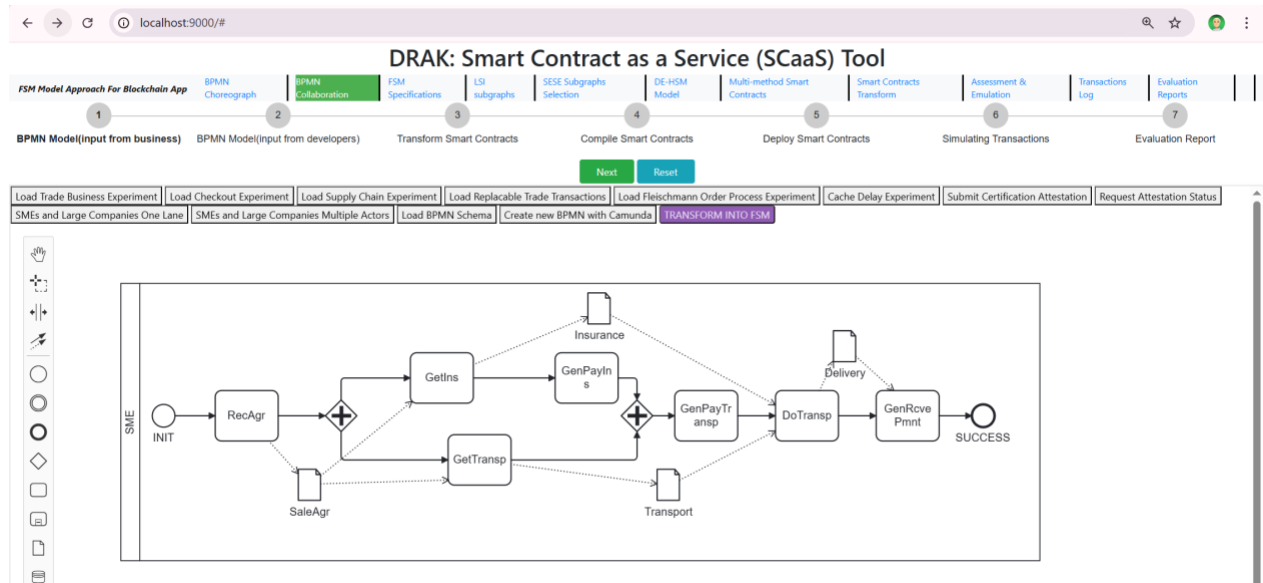


Figure 3. Screenshot of the SCaaS tool preparing the BPMN model for the sample use case

After the BPMN model is created, the developer guides its transformation into smart contracts by interacting with the tool and providing implementation code for template methods corresponding to BPMN task elements. The tool supports the generation of smart contracts for both Hyperledger Fabric blockchains and Ethereum Virtual Machine (EVM)-based blockchains, such as Quorum and Ethereum. The generated main-chain smart contract can invoke methods of smart contracts deployed on a sidechain. ***** END - From Tabs+R Elsevier J2 paper *** ***

After the required payment smart contract services have been prepared by the software developer, the BPMN model of the trade activity containing the three generic payment tasks is constructed, as shown in Figure 3. This model is then transformed into a smart contract using the SCaaS tool. The transformation process is guided by the user through a sequence of horizontal tabs displayed in the interface. After the user initiates the transformation, the BPMN model is first converted into a directed-graph representation of a DE-HSM model. This model is analyzed to identify fragments suitable for treatment as transactions, as described in (Liu, 2024a).

Next, for each generic payment task, the tool prompts the modeler to select which of the available deployed payment services should be used to realize that task. In the sample use case, there are three generic payment tasks: two payable payments and one receivable payment. **Figure 4** illustrates the selection process. In the right-hand pane, the modeler first selects a generic payment task and then selects the corresponding payment service to be used. The figure shows that there are three generic payment services listed horizontally, *GenPayIns*, *GenPayTransp*, and *GenRcvdPmnt*, and that the user has chosen the *GenRcvdPmnt* service (button next to it has been highlighted) for selection of the payment service that is to be deployed for the generic service. Below the horizontal list of the generic payment services, the list of payment services, developed and deployed by the software developer, is shown in a vertical column style, with each payment service having a selection button next to it. The listed payment services are *RcveCrypto-MnCh*, *USDXCrypto-payStCn*, *USDYCrypto-payStCn*, and *BankX-payOffCh*. Furthermore, the figure also shows that for the chosen *GenRcvdPmnt* service, the modeler has selected the *RcveCrypto-MnCh* payment service.

As shown on the diagram in a vertical list of available payment services, the following services were implemented and deployed:

- *RcveCrypto-MnCh* ... On main chain payment receivable service
- *PayUSDXCrypto-StCn* ... 3rd-party payable service on the *USDXCrypto-StCn* stable-coin blockchain
- *PayUSDYCrypto-StCn* ... 3rd-party payable service on the *USDYCrypto-StCn* stable-coin blockchain
- *PayBankX-OffCh* ... Off chain bank receivable service

The listed payment services are doubles, i.e., they are properly invoked, however, implemented payable services are mocks that are used to test and verify the behavior and interactions between the trade activity and payments.

To the left of the list of generic services and the payment services available, there is a list of the selections made for the generic services. It shows that the following selections that were made:

- *GenPayIns* ... *USDxCrypto-StCn* -3rd-party payable service via the *USDxCrypto-StCn* blockchain
- *GenPayTransp* ... *BankX-OffC* - Off chain bank receivable service
- *GenRcvePmnt* ... On main chain payment receivable service

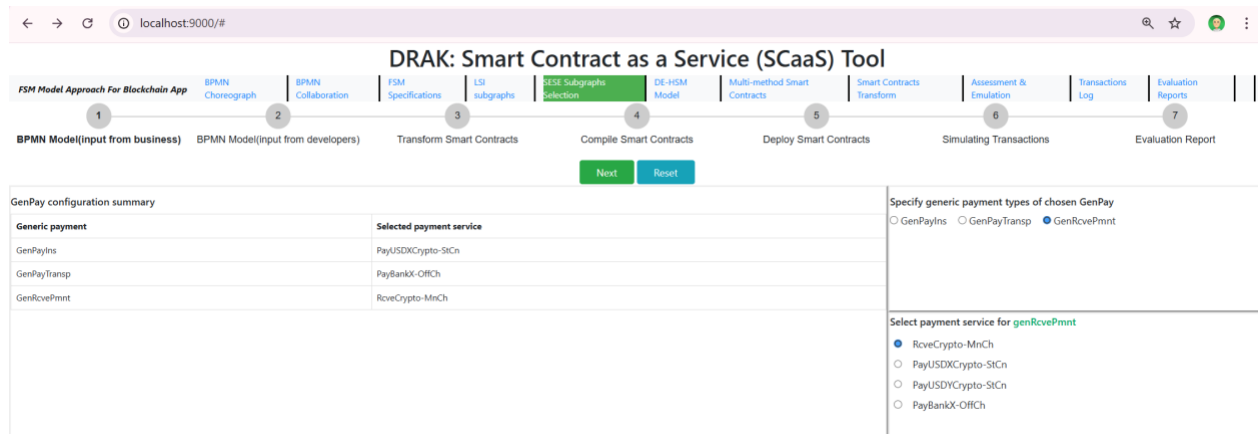


Figure 4. SCaaS tool Showing Selection of Payment Services for Generic Payment Tasks

For each selected payment service, the SCaaS tool verifies that all required information flowing into the BPMN task element is present. If required data is missing, the tool notifies the modeler, who must either amend the BPMN model to include the missing information or select a different payment service. If none of the existing services are applicable, a new payment service must be created by a software developer.

After the generic payment services are mapped to the payment services to be used, the tool performs the transformation of the BPMN model into the smart contract methods and API, and the trade-activity smart contract is deployed on the target blockchain. In addition, the smart contract is deployed on the target blockchain. The tool supports using two blockchain types: (i) Hyperledger fabric and (ii) Ethereum or EVM-based blockchain such as Quorum. Either type can be used for the mainchain or for a sidechain(s), wherein the mainchain smart contract invokes methods of the smart contract deployed on a sidechain.

Recall that payment services smart contracts are created by the software developer. Also recall that the transformation method prepares special API methods that listen to the events, emitted by the smart contract as log records, that indicate the progress of the trading activity. The log messages are used by the API to determine the state of execution of the smart contract and present it to the user.

3.5.4 Execution

Once the transformation of the BPMN model results in the deployment of the smart contract for the trade activity and the APIs are instantiated, execution is controlled by the SCaaS tool that has been built for research experimentation and evaluation purposes with the objective to achieve a proof-of-concept. The tool thus provides functionalities targeted to the development of software and its evaluation rather than execution in production environment. The modeler can examine the generated system execution by stepping through execution message-by-message, while the tool shows the progress graphically by showing the change to states of the individual FSMs representing the DE-FSM sub-models. This feature is helpful in testing and manual verification. Delays are also shown as execution proceeds step-by-step. The tool also provides for bulk issuance of transactions, so that the modeler can instruct the tool to measure average execution delays when trading activities are issued with a particular frequency so that execution delays properties may be measured.

Figure 5 shows a screenshot of the tool at the beginning of the execution. At the top of the screen, underneath the title, there are horizontal tabs that control the different phases and sub-phases of the transformation. Below it are buttons for controlling the execution, namely *Next* and *Reset*, and *Start Evaluation* and *Stop Evaluation*. To the left of the *Start Evaluation* button, there is a selection for either (i) load testing, when many trade activities are executed

to derive averages such as average delays, or (ii) stepping through the execution to view details when testing the software using manual verification.

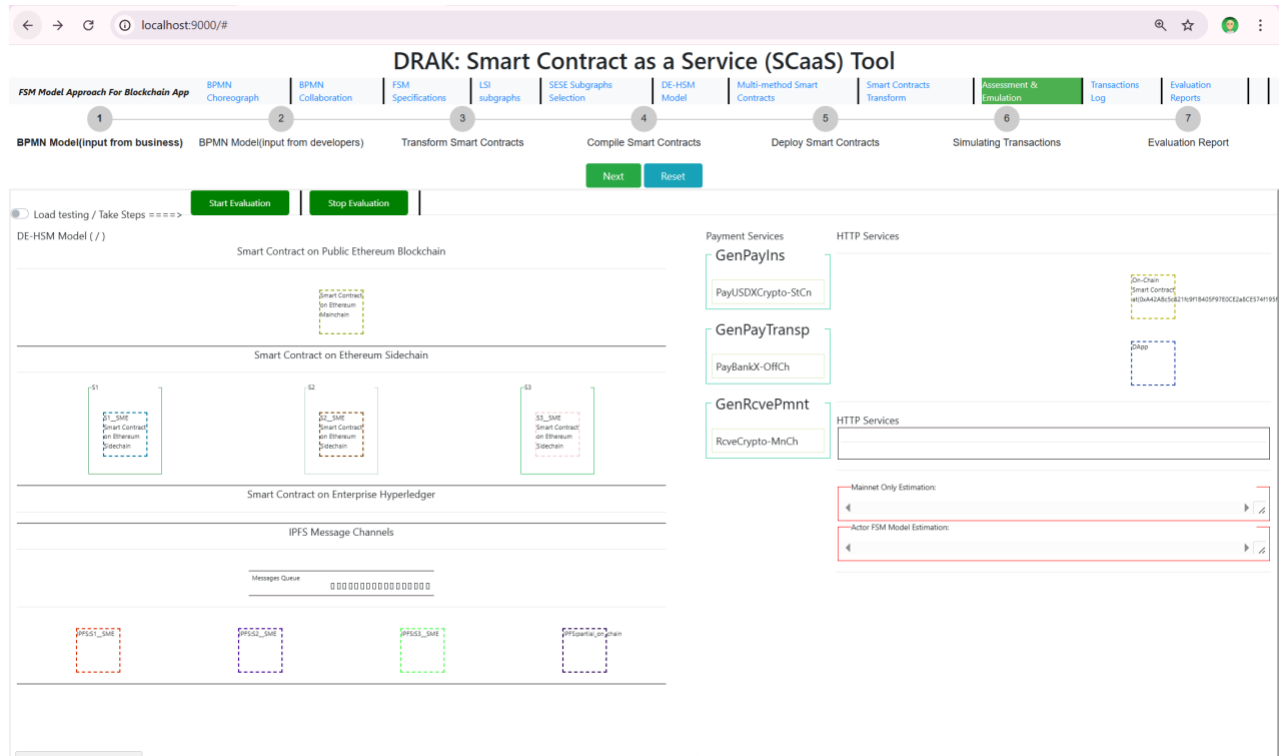


Figure 5. SCaaS Tool – Screenshot of SCaaS for controlling and observing execution

Below the Start Evaluation and Stop Evaluation buttons and the Load-testing or Step-through options, on the left-hand side, are shown the main components of the DE-HSM model as rectangles—namely, the smart contracts for the main trading activity, the three payment methods, and the monitoring smart contract (Bodorik, Liu, Jutla, 2023). In the middle, we show the selected payment service, wrapped by the corresponding generic payment service that is identified from the BPMN model. Also shown is information about communication with IPFS, which is used to store all documents. Below this is information about the invocation of HTTP services, such as communication with external APIs or with IPFS for storage services.

Recall that the smart contracts emit event messages that are collected by a dedicated API. These messages provide detailed information about changes in the execution state and are used for multiple purposes, including manual verification, debugging, and the collection of performance metrics (e.g., execution delays) for evaluation. In general, the event messages convey the current execution status and maintain a log of execution progress.

Our long-term plan is to use these messages to monitor the execution of trade activities. Specifically, the messages will enable reverse mapping from the execution model back to the BPMN model and will be presented to the operator in graphical form. In this visualization, BPMN elements that have been successfully executed will be shown in green, currently executing tasks in orange or yellow, and tasks that have not yet been executed in gray. Failed activities will be highlighted in red.

At present, the event messages emitted by the smart contracts are displayed in a separate window as they are received. **Figure 6** shows this event-message window in which event messages are organized into four columns. The first column shows the name of the executed task, or BPMN element. The second column lists the tasks that may be executed next upon completion of the current task. The third column displays the parameters used by the task, including the data that flows into the corresponding BPMN task element via incoming sequence flows. The fourth column presents the task result, i.e., the content of the transaction as recorded on the blockchain used for the payment service.

The figure captures the flow of control in the BPMN model as it occurred during execution. The event messages indicate that the execution reached the *RecAgr* task, followed by a parallel split gateway. From this gateway, the flow of control is divided into two concurrent execution paths: one path consists of two sequential tasks, *GetIns* and

GenPayIns, while the other path consists of the *GetTransp* task. This concurrent execution results in the sequence of event messages received from the tasks *GetIns*, then *GetTransp*, and finally *GenPayIns*.

The flow of control from both concurrent paths subsequently reaches the exclusive join gateway that allows the flow to proceed only after both execution paths have completed (i.e., in BPMN terminology, when tokens from both incoming sequence flows arrive at the gateway). However, in our implementation of the transformation from a BPMN model to smart contracts, the join gateway is not explicitly implemented. Instead, we channel the incoming sequence flows to the gateway directly to the task that follows the join gateway, the *GenPayTransp* task. Of course, the *GenPayTransp* task execution starts only if the control tokens arrive on all incoming sequence flows to it.

As indicated in Figure 6, once the execution of the *GenPayTransp* task is completed, as expected, the remaining tasks *DoTransp*, *GenRcvPmnt*, and *Success* are executed as confirmed by event messages shown in the figure.

Execution of the payment services is achieved by invocation of the payment services prepared by the software developer. For each generic payment, the SCaaS transformation from BPMN model causes invocation of a method of a smart contract prepared by the software developer. The invocation of the payment service is facilitated with the use of a multi-step transaction supported by SCaaS. The payment service is treated as a multi-step transaction and the generic payment task is just a mechanism for its invocation.

new_state	next_new_state	input	output
INIT	RecAgr	"N/A"	"N/A"
RecAgr	Gateway_Olg76nx	"initialization"	{ "productId": "product_identification", "maximumPrice": 100, "nextState": "GetTrReq" }
Gateway_Olg76nx	GetIns,GetTransp	"SaleAgr"	"N/A"
GetIns	GenPayIns	{ "productId": "product_identification", "maximumPrice": 100, "price": 98 }	Insurance
GetTransp	GenPayTransp	{ "productId": "product_identification", "maximumPrice": 100, "price": 98 }	Transport
GenPayIns	GenPayTransp	{ "fn": "Pay", "payer": "0x3a642c4ce0735549868cf1f0f153cdf2b6b58b84", "payee": "0xffff6a862df7c85b5d20f08bccf6c95ba270d7f93", "amount": 1000000000000000000 }	{ "txHash": "0x77ba980ecfacf2032e869844876c45e1804820713490233e63d58fe1e5b44440", "status": true, "blockNumber": 1007246, "gasUsed": 54499, "event": "Pay", "payer": "0x3a642c4ce0735549868cf1f0f153cdf2b6b58b84", "payee": "0xffff6a862df7c85b5d20f08bccf6c95ba270d7f93", "amount": 1000000000000000000 }
GenPayTransp	DoTransp	{ "fn": "Pay", "payer": "0x3a642c4ce0735549868cf1f0f153cdf2b6b58b84", "payee": "0x12039f15bd079bb2220bda47b2df6d379c583fad", "amount": 1000000000000000000 }	{ "txHash": "0xe9f20f89586c21b02515fb3bd8b2070690230c02b2834b1efb9b54ae97e93f4d", "status": true, "blockNumber": 1003300, "gasUsed": 65846, "event": "Pay", "payer": "0x3a642c4ce0735549868cf1f0f153cdf2b6b58b84", "payee": "0x12039f15bd079bb2220bda47b2df6d379c583fad", "amount": 1000000000000000000 }
DoTransp	GenRcvPmnt	Insurance, Transport	Delivery
GenRcvPmnt	SUCCESS	{ "fn": "Pay", "payer": "0x5af29518e5171df3d7dc3ec27067a22c165081cb", "payee": "0xea43ed7825a662f647f12198485cc2da0ef63f1c", "amount": 1000000000000000000 }	{ "txHash": "0x630338ae14081081507bf62c258071871dc62fa011928f0dbb365b6129f43e18", "status": true, "blockNumber": 1082724, "gasUsed": 43857, "event": "Pay", "payer": "0x5af29518e5171df3d7dc3ec27067a22c165081cb", "payee": "0xea43ed7825a662f647f12198485cc2da0ef63f1c", "amount": 1000000000000000000 }
SUCCESS	N/A	Delivery Confirmation	"N/A"

Figure 6. Window with Event Messages

4. Re-use and Extensibility with Settlement-netting Payment Services

In this section we focus on re-use and extensibility by creating a payment service that performs *settlement with netting*, also referred to as *bulk payments with netting*. Re-use allows the same payment service to be used in many trade activities, while extensibility allows creation of a new service while reusing an already existing service, without altering the existing service. We chose the bulk payment with netting service because it requires interaction between parties, interaction that shares the common state, and it lends itself for re-use of existing payment services.

We first discuss the settlement with netting method, then present and discuss the BPMN model for the trade activity, describe the payment services in terms of its BPMN model and its transformation into a smart contract payment services, and finally we discuss the execution of the blockchain application.

4.1 Settlement with Netting aka Bulk Payments with Netting

Settlement with netting is used by two trading partners in situations when they have many trade transactions with payments in both directions. To minimize the overhead cost due to services charged for each payment, the parties agree to use settlement netting that aggregates the amount due among for each of the parties and nets the cash flows into one payment. In other words, only the net difference in the aggregate amounts is delivered or exchanged by the party with the net owed obligation. Typically, an agreement for the process of settlement netting must be in place, such that:

- All payments due between participants are grouped together (bulk).
- Amounts owed to and from each participant are offset (netting).
- Conditions when netting payments are made are specified.
- Each participant pays or receives one net amount instead of many separate payments.

This approach reduces the number of payments, lowers settlement risk, and improves efficiency in systems like clearing houses, banking, and financial markets (Kenton, 2025). The payment netting agreement must indicate conditions when the settlement payments for the net positions are made, such as the settlement is made at the end of each day; when the net position exceeds certain threshold limit; after certain number of payments are made; or when some other conditions are met.

As the use case shown in [Figure 1](#) does not lend itself for the settlement with netting, we created a simple use-case, shown in [Figure 7](#), for a participant making payments that use settlement with netting. For simplicity, a participant receives an approved invoice to be paid. After the payment is made, the participant performs local recording of the result. For simplicity, there are only two participants, X and Y, and each one uses the same BPMN model.

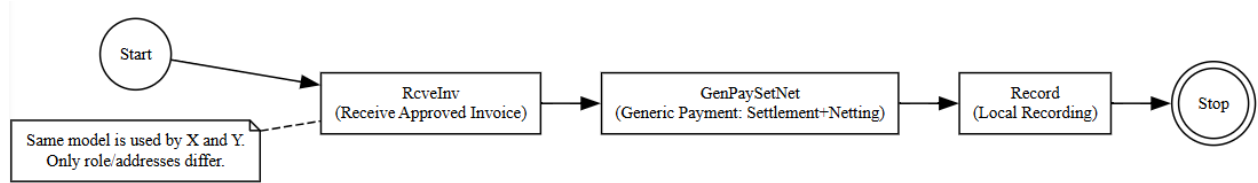


Figure 7. BPMN model for the Settlement with netting use case

4.2 Payment Service for Settlements with netting

The payment service that uses settlement with paying will use an already existing payment service *StCnPayXY* to facilitate payments between X and Y. We first overview this method and then we proceed with a brief description of the settlement with netting method and then describe creation of the payment service that uses settlement with netting.

4.2.1 Payments between X and Y

The *StCnPayXY* payment facilitates transfer of currency between X and Y. It was created as per description in Section 3 and then deployed on a 3rd-party stable-coin blockchain, say *StCnBC*, using its crypto currency *CryptoCtCn* pegged to some fiat currency. For instance, *StCnBC* might be *USD Coin (Tether)* blockchain with its currency *USDC (UDTH)*. The parameters for the *StCnXY* method include account numbers for X and Y, number of stable coins to transfer between X and Y, and direction of payment that identifies which of X and Y is a payee and a payer. Of course, information about the payment service is also stored in the payment repository

4.2.2 Payment service that uses settlement with netting

Before creating the payment service, the participants must first agree on the settlement with netting details. To simplify exposition, we assume that that payments must be made if a net position for a participant exceeds the threshold value *NetLimit*, or if the total number of payments made since the last settlement exceeds the threshold value *NumLimit*.

The BPMN model for the payment services is shown in [Figure 8](#), in which part (a) shows the BPMN model, and then parts (b), (c), and (d) show the code for initialization, calculating the net positions, and making the payment, respectively.

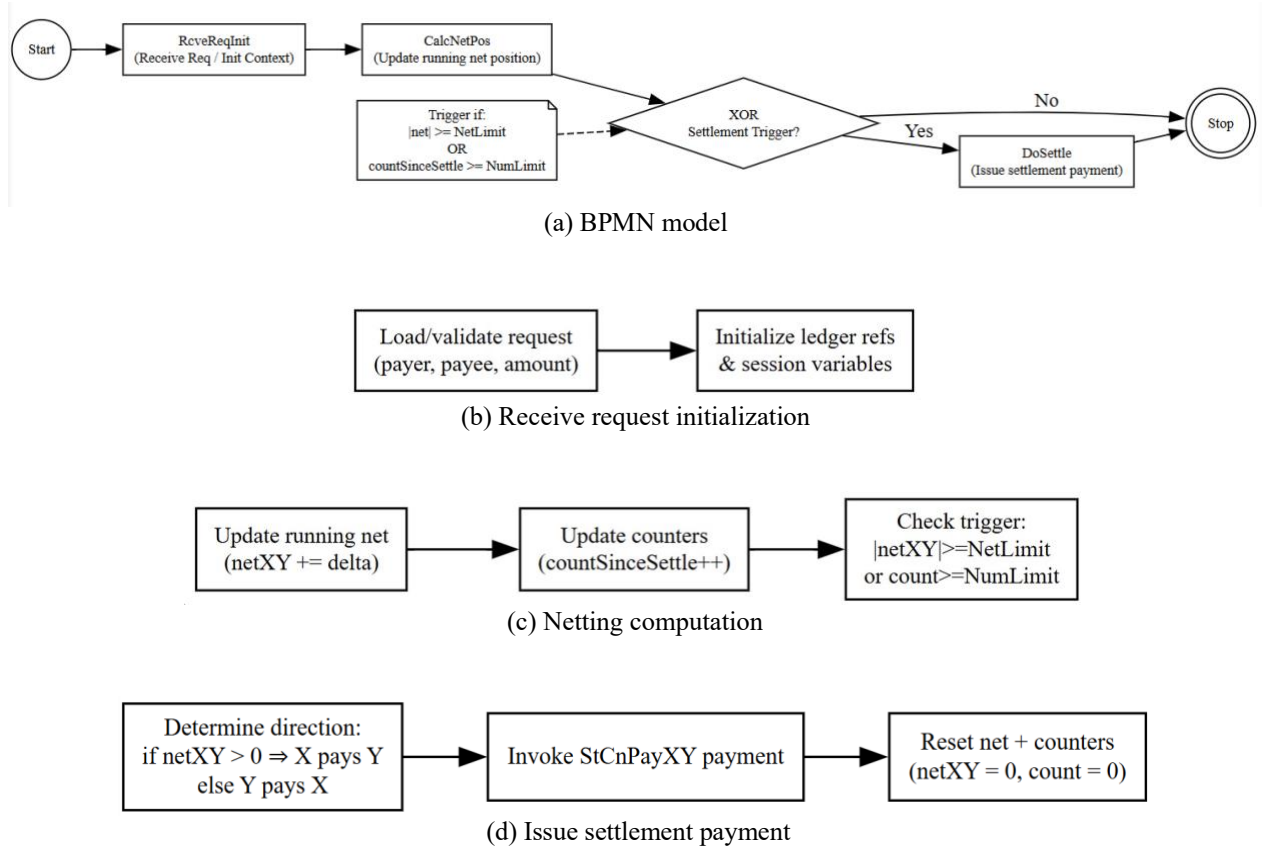


Figure 8. BPMN model and implementation for settlement with netting

The SCaaS tool is used by the software developer to create the payment service, *PaySetNet*, which uses settlement with netting, as was described in the Section 3. BPMN model is created first, which is then transformed using the SCaaS tool into a smart contract. In addition, the payment service information is stored in the repository of payment methods for the *StCnBC* blockchain. The payment service is deployed on the blockchain *StCnBC* if the payment service is invoked cross-chain, then the payment service must also have a helper smart contract the provides a bridge for interaction between the *StCnBC* blockchain and the blockchain from which the request for payment originates.

4.3 Trading activity smart contract generation

In the following, we shall describe how a trading activity that uses the payment service payment service is created and deployed. The BPMN model of the trading activity for X is shown in [Figure 7](#), in which the generic payment *GenPaySetNet* needs to be realized using the payment service *PaySetNet*, which is deployed on the blockchain *StCnBC*. Before we can instantiate the *GenPaySetNet* service, if the information about the *PaySetNet* service is not in the repository of payment services on X, it must be *imported*, and it must be ensured that a helper smart contract exists on the *XpBP* chain to collaborate with the helper smart contract on the payment chain *StCnPayBC* in order to provide a bridge for interaction between the two chains.

Once the payment repository for *XpBC* contains information on the payment service *PaySetNet*, generation of the smart contract and APIs using the SCaaS tool proceeds as described in Section 3. During the transformation, the modeler selects the *PaySetNet* payment service from the repository as instantiation for the generic *GenPaySetNet* payment, and the transformation produces the smart contract and API. [Figure 9](#), which shows that the developer chose that the *PaySetNet* payment service should be used to instantiate the *GenPaySetNet* generic payment service.

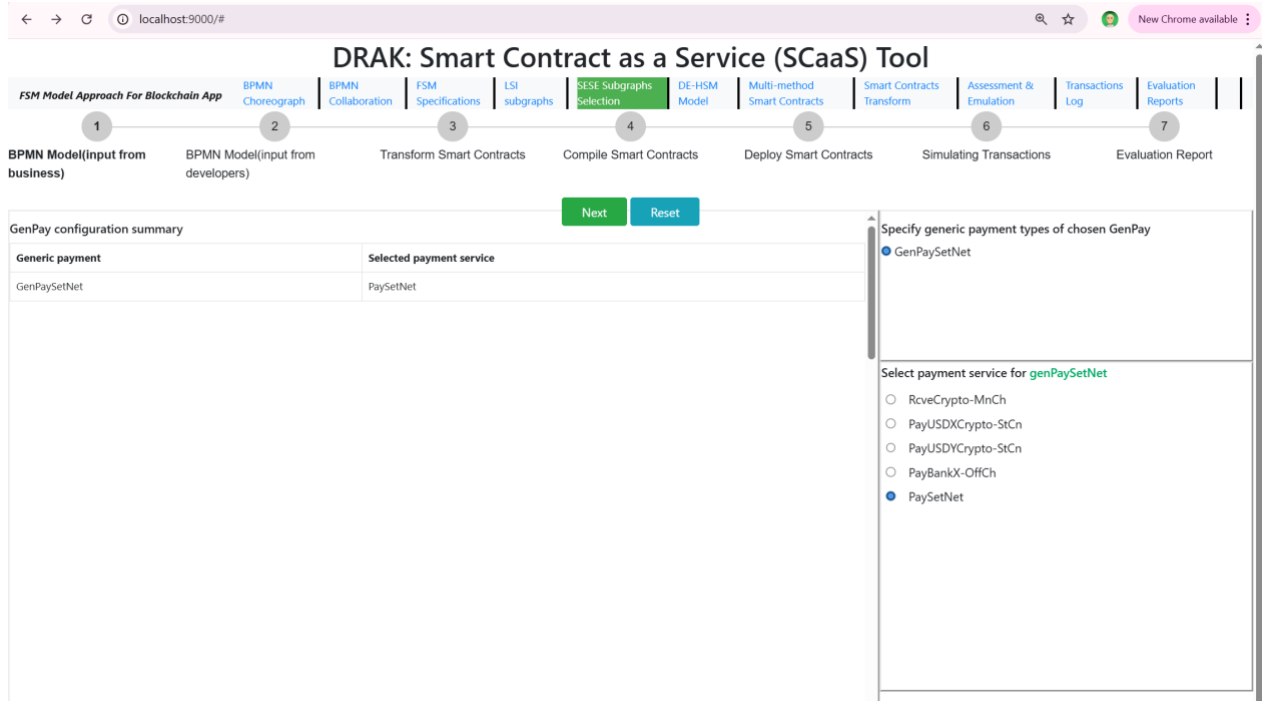


Figure 9. Selection of the actual payment service to be used for the *GenPaySetNet*

Generation of the smart contract and the API by partner Y for the model shown in Figure 7 follows the same process as for the partner X.

4.4 Execution

Recall that we described the execution environment in subsection 3.5.2. We created smart contracts for partners X and Y, as well as a smart contract implementing the payment service.

The smart contracts for Partner X and Partner Y were deployed on blockchains *XpBC* and *YpBC*, respectively, where *XpBC* is an Ethereum-based blockchain, and *YpBC* is a Hyperledger Fabric (HLF)–based blockchain.

The payment smart contract *PaySetNet*, which is responsible for maintaining the net position and determining when settlement should be triggered, was deployed on the Ethereum-based payment service chain, *StCnBC*.

Also recall that the payment service *StCnPayXY*, which is used to perform the actual transfer of stablecoins between partners X and Y upon settlement, is deployed on the same payment service chain, *StCnBC*, and is invoked via the payment helper bridge smart contracts.

Furthermore, for simplicity, we chose $NetNum = 2$, so we could observe payments being made. We chose $NetLimit$ to be arbitrarily 100. We chose step-by-step execution for testing and used different values for $NetNum$ and $NetLimit$ to ensure that netting is performed correctly and payments are issued appropriately.

Figure 10 shows architectural diagram of the software components involved when the payment services is invoked by the trading activity smart contracts for X and Y.

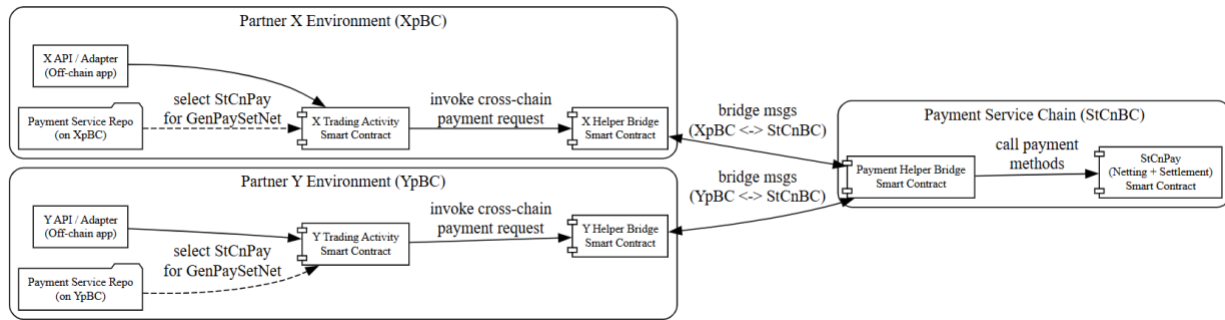


Figure 10. Architectural diagram of the software components interaction to realize the payment service

As indicated before, we set the threshold value *NetNum* = 2 and chose bulk testing by issuing multiple transactions. We should observe that after two payments are invoked, a payment should be performed by transfer of funds between X and Y to achieve zero net position. **Figure 11** shows a window containing event messages generated while executing the deployed payment smart contract.

Real-time Logs - Google Chrome

about:blank

Real-time Logs (Settlement + Netting)

new_state	next_new_state	input	output	associatedDocument
INIT	RcvInv	"N/A"	"N/A"	N/A
RcvInv	GenPaySetNet	{ "invoiceId": "INV-0001", "from": "X", "to": "Y", "amount": 60 }	{ "status": true }	Approved Invoice
GenPaySetNet	StCnPay RcvReqInit	{ "payer": "X", "payee": "Y", "amount": 60, "agreement": { "netLimit": 100, "numLimit": 2 } }	{ "event": "RcvReqInit", "requestId": "REQ-1001", "netBefore": 0, "countBefore": 0 }	Netting Agreement
StCnPay CalcNetPos	SplitGate	{ "requestId": "REQ-1001", "delta": 60 }	{ "netAfter": 60, "count": 1, "trigger": false, "reason": "Num<2 and net <60" }	N/A
RcvInv	GenPaySetNet	{ "invoiceId": "INV-0002", "from": "Y", "to": "X", "amount": 20 }	{ "status": true }	Approved Invoice
GenPaySetNet	StCnPay RcvReqInit	{ "payer": "Y", "payee": "X", "amount": 20, "agreement": { "netLimit": 100, "numLimit": 2 } }	{ "event": "RcvReqInit", "requestId": "REQ-1002", "netBefore": 60, "countBefore": 1 }	Netting Agreement
StCnPay CalcNetPos	IssueSettlement	{ "requestId": "REQ-1002", "delta": -20 }	{ "netAfter": 40, "count": 2, "trigger": true, "reason": "NumLimit reached (2)" }	N/A
IssueSettlement	PaySetNet	{ "netPosition": 40, "directionRule": "net=0 => X pays Y; net=0 => Y pays X", "lastKnownHoles": [{ "party": "X", "address": "0x3a642c4ce73549868cf1f0f153cf2b05b084", "token": "USDC" }, { "party": "Y", "address": "0xffff6a82df7c85b5d2f08bcccfc95ba27b07f93", "token": "USDC" }, "derivedFromRequests": ["REQ-1001", "REQ-1002"] }	{ "settlementDecision": true, "payer": "0x3a642c4ce73549868cf1f0f153cf2b05b084", "payee": "0xffff6a82df7c85b5d2f08bcccfc95ba27b07f93", "amount": "40", "nextAction": "Call PaySetNet() on USD contract" }	N/A
PaySetNet	ResetNet	{ "for": "transfer", "token": "USDC", "tokenContract": "0xa0b86991c6218b36c1d19d4a2e9eb0cc3086e48", "payer": "0x3a642c4ce73549868cf1f0f153cf2b05b084", "payee": "0xffff6a82df7c85b5d2f08bcccfc95ba27b07f93", "amount": "40" }	{ "transactionHash": "0x77b9b88ecfac2802e60844876c45e184820713490233eb3d5f8e1a5b4448", "blockNumber": "0xf5a3b", "status": "Bvt", "true": "0x3a642c4ce73549868cf1f0f153cf2b05b084", "to": "0xa0b86991c6218b36c1d19d4a2e9eb0cc3086e48", "gasPrice": "0x0f2b", "type": 0, "event": "Transfer", "token": "USDC", "from": "0x3a642c4ce73549868cf1f0f153cf2b05b084", "to": "0xffff6a82df7c85b5d2f08bcccfc95ba27b07f93", "value": "40" }	ERIC-20 Transfer Receipt
ResetNet	Record	{ "netBeforeReset": 40, "countBeforeReset": 2, "settlementTx": "0x77b9b88ecfac2802e608448" }	{ "netAfterReset": 0, "countAfterReset": 0, "note": "Reset after successful settlement transfer" }	N/A
Record	Stop	{ "localRef": "LOCAL-7788" }	{ "status": true }	Local Record
Stop	N/A	"N/A"	"N/A"	N/A

Scenario: NetLimit = 100, NumLimit = 2. Two requests accumulate net = +40, trigger settlement. PaySetNet transfers USDC and ResetNet clears net state.

Figure 11 Event messages emitted

***** What follows is still to be re-written amended *****

After the initialization represented by the INIT event, an event message is received from the execution of the *RcvInv* task by the partner X, which is followed by the execution of the *GenPaySetNet* task. As a result, the payment services is invoked. As a result, there is interaction between the helper smart contracts, wherein the helper smarty contract deployed on the payment chain, *StCnBC*, validates the request and performs initialization of the payment service. As this is performed by the helper smart contract, there is no event message that is emitted that

corresponds to the *RcveReqInit* task shown in Figure 8. Thus, the next event message received is from the task *StCnPay.CalcNetPos*. As settlement payment is not required, execution of the payment service completes.

Next two event messages are similar to those of X: An event message is received from the execution of the task *RcveInv* by the partner Y, which is followed by an event message from the task *GenPaySetNet* to request a payment to be made by Y. The helper smart contracts on *YpBC* and *StCnBC* chains interact, with the helper smart contract on the chain *StCnBC* performing the request validation, initialization, and then causing execution of the task *StCnPay.CalcNetPos*. However, as this is the second payment request, after calculating the net position, the payment is issued followed by a task to reset net position. Finally, the last task, *Record*, of the payment service is executed.

5. Related Work

We first provide literature related to our approach to the generation of smart contracts from BPMN models and multi-step transactions (Bodorik, Liu, Jutla, 2023; Liu, Bodorik, Jutla, 2024), and then we provide literature related to work in this paper on blockchain-based payment processing.

5.1 Generation of Smart Contracts from BPMN Model

Several research initiatives have explored transforming Business Process Model and Notation (BPMN) models into executable smart contracts. The Lorikeet project employs a two-phase transformation approach (Tran et al., 2018). In the initial phase, the BPMN model undergoes analysis and transformation into smart contract methods, which are subsequently deployed on the Ethereum blockchain. An off-chain component manages communication between the decentralized application (Dapp) and external actors. Participants exchange messages according to the BPMN choreography, with an off-chain component coordinating these exchanges. The smart contract incorporates a monitoring mechanism that records and enforces the choreography model, ensuring message exchanges follow the predetermined sequence. Additionally, Lorikeet supports asset management capabilities, including both fungible and non-fungible tokens, providing a registry and methods for asset operations such as transfers. This functionality enables rapid prototyping of smart contracts from BPMN models requiring asset management features, facilitating iterative development, testing, and refinement before final deployment.

The Caterpillar framework presents an alternative methodology, concentrating on BPMN models within a single pool, which is a BPMN structural construct, where all business processes are recorded on the blockchain (López-Pintado et al., 2019). Its architecture comprises three distinct layers: Web Portal, Off-chain Runtime, and On-chain Runtime. The On-chain Runtime layer encompasses smart contracts responsible for workflow control, interaction management, configuration, and process oversight, with Ethereum serving as the primary blockchain platform.

Loukil et al. (2021) introduced the Collaborative Business Process (CoBuP) architecture for executing collaborative business processes on blockchain. Unlike approaches that directly compile BPMN models into smart contracts, CoBuP deploys a generic smart contract that invokes predefined functions. The architecture features three layers: Conceptual, Data, and Flow. BPMN models are transformed into a JSON Workflow representation that governs process instance execution and manages interactions with blockchain data structures.

Di Ciccio et al. (2019) conducted a comparative analysis of the Lorikeet and Caterpillar approaches across multiple dimensions, including model execution capabilities, BPMN element coverage, detection of incorrect behaviors, sequence enforcement mechanisms, participant selection, access control, and asset management. Their analysis highlighted the distinctive characteristics of each approach and established a foundation for systematic comparison.

Our approach to generation of smart contracts from BPMN models distinguishes itself through the transformation of BPMN models into Directed Acyclic Graph (DAG) representations, followed by conversion to Discrete Event Hierarchical State Machine (DE-HSM) and Discrete Event Finite State Machine (DE-FSM) models. This transformation enables systematic analysis of process flows to identify localized processing using Single Entry Single Exit (SESE) subgraphs. In contrast to systems that directly translate BPMN models into smart contract code, TABS+ generates smart contracts as abstract representations of process flows, with implementation details expressed through concurrent FSMs. The process logic executes through a smart contract that manages FSM state transitions, with specific transitions triggering the execution of localized BPMN tasks. Each task is implemented as a smart contract method with clearly defined inputs and outputs. Furthermore, localized SESE subgraphs can be packaged and deployed as independent smart contracts, potentially on sidechains for improved efficiency, or utilized to define nested trade transactions with formally specified transactional properties.

To the best of our knowledge, no prior formal work has addressed multi-step, multi-method transactions for blockchain smart contracts, what we term trade transactions. These multi-step transactions contain executions of multiple independently invoked smart contract methods by different actors.

5.2 Smart Contract Upgradeability and Repair

The immutability property of blockchain technology, while promoting trust, creates challenges when collaborative activities require modifications to repair security issues or incorporate new features. Smart contract upgradeability has attracted substantial research attention, as evidenced by comprehensive literature surveys on the topic (Meisami & Bodell, 2023; Girish & Kaganurmath, 2022; Palladino, 2018). Several bug fixing frameworks, e.g., EVMPatch (Rodler et al., (2021), SolSaviour (Li et al., 2021), FlexChain (Corradini et. al, 2020, 2022), and BlockME (Falazi et al., 2019).

The work most closely aligned with our repair approach is that of Klinger et al. (2020), which analyzes and implements three upgradeability concepts: a registry-based approach, a proxy pattern, and a combined registry with pattern segregation. Their use case involves a large organization with multiple departments, where the BPMN model of collaboration transforms into smart contracts, with each department's activities represented by a separate contract. Upgrade management must ensure consistency between previously executed activities and the upgrade context. Their findings indicate that the Unstructured Storage Proxy pattern shows the most promise for practical deployment, particularly regarding cost-effectiveness and minimal added complexity.

Compared to Klinger et al. (2020), the TABS+R approach naturally packages activities into separate smart contracts, exploiting nested trade transactions to protect against inconsistencies arising when some activities complete using the old contract version while others execute with the newly repaired or upgraded contract.

5.3 Payment Rails for Smart Contracts in Blockchain Applications

Our work on payment rails for smart contracts builds upon the above research trajectory and intersects more specifically with literature on blockchain-based payment processing, settlement mechanisms, and financial workflows modeled as business processes.

A substantial body of research explores blockchain-enabled payment systems and settlement infrastructures, often focusing on performance, trust minimization, or cross-organizational coordination. However, many such systems are designed directly at the protocol or smart-contract code level, without leveraging high-level business process models. As a result, the alignment between business requirements and deployed contract logic remains difficult to verify or evolve.

Within the BPMN-to-smart-contract literature, most prior work treats payments as simple task executions or token transfers embedded within a broader workflow (López-Pintado et al., 2018; Jin et al., 2024). These approaches typically lack explicit modeling of payment-specific concerns such as settlement modes, batching, netting, or conditional execution based on financial thresholds.

The approach to blockchain-enabled payments extends existing BPMN-driven smart-contract generation by explicitly focusing on payment orchestration and settlement logic as first-class process constructs. In doing so, it complements our earlier SCaaS/TABS-based work (Bodorik, Liu, Jutla, 2023; Liu, Bodorik, Jutla, 2024; & Liu, Bodorik, Jutla, 2025) by refining how payment-related BPMN patterns are identified, transformed, and executed on blockchain platforms. This focus aligns with emerging research on blockchain-based financial infrastructure, while retaining the model-driven advantages of our approach to transformation of BPMN models to smart contracts.

Furthermore, our work distinguishes itself from generic payment-contract frameworks by integrating payment logic with transactional guarantees and lifecycle management mechanisms inherited from TABS+ and TABS+R. This integration allows payment processes to be executed, monitored, repaired, or upgraded in a manner consistent with the original BPMN specification, addressing limitations of static or hard-coded payment smart contracts. In this sense, our work on payment rails occupies a niche at the intersection of BPMN-based smart-contract generation and blockchain payment systems.

6. Summary and Conclusions

6.1 Summary

This paper addressed the problem of systematically integrating payment rails into Smart-Contract-as-a-Service (SCaaS) solutions generated from BPMN models. While prior work by the authors established a model-driven approach for transforming BPMN business processes into executable smart contracts, payment handling in those solutions remained largely implicit or implementation-specific. Given the central role of payments in inter-organizational processes, supply chains, and financial workflows, the lack of explicit modeling and transformation support for payment mechanisms represents a significant gap.

The primary objective of this research was to define, model, and operationalize payment rails within BPMN-to-smart-contract transformations. To achieve this objective, the paper analyzed common enterprise and financial payment patterns, including real-time gross settlement, deferred net settlement, and bulk or threshold-based settlement, and examined how these patterns can be formally represented at the BPMN level and correctly realized at the smart-contract level.

Building on the SCaaS architecture and transformation pipeline developed in earlier work, the paper proposed extensions to BPMN modeling conventions, execution semantics, and smart-contract generation logic to explicitly capture payment behavior. The resulting approach ensures that payment execution, settlement, reconciliation, and netting semantics are preserved across the model-to-code boundary, while remaining adaptable to different blockchain platforms and off-chain payment infrastructures.

The key contributions of this paper, expressed in more general terms in comparison to the specific contributions listed in Section 1, are as follows:

- *Conceptualization of Payment Rails for SCaaS* - the paper introduces a structured classification of payment rails relevant to smart-contract-based business processes and formalizes their operational characteristics in a way that is compatible with BPMN modeling and blockchain execution.
- *BPMN Modeling Extensions for Payments* - The work demonstrates how payment-related concerns—such as settlement mode, batching, thresholds, and reconciliation—can be explicitly represented in BPMN models using well-defined patterns, annotations, and subprocess structures.
- *Model-Driven Transformation to Smart Contracts* - The paper extends the existing BPMN-to-smart-contract transformation framework to support payment rails, ensuring that payment logic is generated consistently, traceably, and in alignment with the modeled business intent.
- *Integration of On-Chain and Off-Chain Payment Semantics* - The proposed approach bridges on-chain smart-contract execution with off-chain payment systems, providing a practical foundation for enterprise adoption where blockchain does not natively provide all required payment functionalities.
- *Demonstration Through Representative Scenarios* - The paper illustrates the applicability of the approach using representative payment scenarios, showing how different payment rails influence process execution, settlement behavior, and contract state transitions.

6.2 Conclusions

This research demonstrates that payment handling in smart-contract-based systems cannot be treated as a secondary or purely technical concern. Instead, payment rails must be explicitly modeled and systematically transformed alongside core business logic to ensure correctness, compliance, and interoperability. By elevating payment rails to first-class citizens in BPMN-driven SCaaS solutions, the proposed approach improves transparency, reduces ambiguity between business intent and technical implementation, and supports a broader range of real-world enterprise use cases.

The results show that a model-driven, payment-aware SCaaS architecture can accommodate diverse settlement mechanisms without sacrificing automation or scalability. Moreover, the separation of concerns between business process modeling, payment semantics, and execution infrastructure enables greater flexibility in adapting solutions to evolving regulatory, technological, and organizational requirements.

Future work will focus on automated compliance checking against financial regulations, and empirical evaluation of the approach in large-scale, multi-party deployments. Additionally, further exploration of interoperability with emerging digital currency infrastructures and cross-chain payment mechanisms represents a promising direction for extending the SCaaS paradigm.

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